

BOTSWANA COLLEGE OF ENGINEERING AND TECHNOLOGY

Department of Key Skills

Course: ICT122 – Computing Skills Fundamentals II

Assignment One

Submission Period: March 10 - March 15

Total Marks: 50

ASSIGNMENT OVERVIEW

In this assignment, you will create a spreadsheet that showcases your ability to use various functions and features of spreadsheet software (e.g., Microsoft Excel, Google Sheets). You will produce a unique piece of work based on your college student ID. The assignment consists of data analysis, visualization, and presentation skills.

ASSIGNMENT TASKS

1. Data Entry (20 Marks)

- Create a new spreadsheet file.
- Use your student ID to generate a dataset. For example, if your student ID is 123456, create a dataset of 12 students with the following attributes:
 - Student ID
 - Name
 - Course
 - Year of Study
 - Marks in 5 different courses you were studying last semester (e.g., ICT121, GEC111, etc.)

Instructions:

- Use the last three digits of your student ID to decide the number of student records to include (e.g., if your ID ends in 456, include 6 students). If the last digit of your student ID is less than 5, use the last 2 digits to determine the number of records to include.
- Ensure that your dataset is realistic and contains varied entries.

2. Calculations (10 Marks)

- Use formulas to calculate:
 - Total Marks for each student.
 - Average Marks for each student.

- Pass/Fail status for each student based on an average cut-off of 50%.

Instructions:

- Use appropriate functions (SUM, AVERAGE, IF) to perform these calculations.
- Ensure your formulas are correctly applied and update automatically if data changes.

3. Data Visualization (10 Marks)

- Create at least two types of charts to visualize the data:
 - A bar chart showing total marks for each student.
 - A pie chart illustrating the percentage of students passing vs. failing.

Instructions:

- Label your charts appropriately and include titles.
- Ensure that the colors and styles used enhance readability.

4. Presentation and Documentation (10 Marks)

- Write a brief summary (200-300 words) explaining your dataset, what you learned from the analysis, and the process of creating your spreadsheet.
- Ensure that your spreadsheet is well-formatted (e.g., clear headings, appropriate cell sizes, consistent font usage).

Instructions:

- Use text boxes or comments for your summary within the spreadsheet.
- Check for spelling and grammatical errors.

SUBMISSION

- Submit your completed spreadsheet through Google Classroom by the end of the submission period (March 15).
- Name your file using the format: "ICT122_StudentID.xlsx" or "ICT122_StudentID.ods".

RUBRIC FOR MARKING

Criteria | Excellent (15-20) | Good (10-15) | Fair (0-9) | Marks

Data Entry

Comprehensive dataset with varied entries and correct attributes.

Dataset is complete but lacks variety or has minor errors.

Incomplete dataset or significant errors in entries. ____ /20

Calculations

All formulas are correct and used effectively.

Most formulas are correct, but some calculations are missing.

Many errors in formulas; calculations are incorrect. ____/10

Data Visualization

Clear, informative charts that effectively communicate the data.

Charts are present but may lack clarity or effectiveness.

Charts are missing or poorly constructed. ____/10

Presentation and Documentation

Well-organized, clear, and free of errors, with a strong summary.

Generally organized but has some errors or lacks clarity.

Poorly organized, unclear, or numerous errors in documentation. ____/20

Total ____ /50

CONCLUSION

This assignment aims to enhance your understanding of spreadsheets and develop your data analysis skills. Ensure to follow each step carefully, and do not hesitate to seek help if needed. Good luck!